

$\mathbb{Z}[\varphi]$ arithmetic in the monstrous atlas: moonshine scaffolding for the Golden Substrate seam

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Abstract

We summarise four pillars—Ogg primes, Conway–Norton Hauptmoduln, Mayer-type generators on $\mathbb{Q}(\sqrt{5})$, and Witten’s $c=24$ holographic conjecture—and record how $\mathbb{Z}[\varphi]$ behaves on the discrete moonshine primes. We excerpt the truncated $T_{11A}(q)$ series as certified in-code and document brute Apollonian norm statistics that fail coefficient-level matching—a necessary negative result, since counting functions (polynomial growth) cannot equal Hauptmoduln (exponential growth) coefficient-by-coefficient. We then show that the generators of $\Gamma_\varphi^{(2)}$ satisfy three group-theoretic conditions fully consistent with the structural conjecture: (i) the minimal polynomial of each generator entry encodes level 11 directly; (ii) the cusp set $\{0, 1, \infty\}$ of Γ_φ matches the cusps of $\Gamma_0(11)$ prior to Atkin–Lehner identification; (iii) cyclic words carry Galois-symmetric traces of the form -11^3 , propagating level-11 arithmetic multiplicatively into the geodesic spectrum. Finally, we reframe the entire landscape using the Hopf line bundle: the modular weight ladder is the Hopf degree ladder over $X(1) \cong \mathbb{CP}^1$, the two spinor chiralities give two strands (Monster–Hecke at integer weight; umbral-pariah at half-integer weight), and the seam at weight one consists of CM forms. At the prime $p = 11$ five independent designations coincide—Ogg prime, E_{11} conductor, $\varphi^5 + \psi^5 = 11$, $11 \mid |\text{O’Nan}|$, and Heegner number $h(\mathbb{Q}(\sqrt{-11})) = 1$ —each separately certified by tests. We restate the conjecture in its correct form: Γ_φ is the real-quadratic geometry that threads this single CM gate, and at the level- $\mathfrak{m} = (11)$ congruence subgroup $\Gamma_0(\mathfrak{m}) \subset \Gamma_\varphi$, the Selberg zeta factors through the base-change Hilbert L-function $L(E_{11}/\mathbb{Q}(\sqrt{5}), s) = L(s, E_{11}) \cdot L(s, E_{11} \otimes \chi_5)$. A direct numerical test of the simpler statement “ $Z_{\Gamma_\varphi} \sim L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ on the full Hilbert modular group” falsifies the naive form: the two decay scales are mismatched by the exact algebraic ratio $\log \varphi^{20} / \log 5 = 20 \log_5 \varphi \approx 5.98$, where φ^{20} is the fundamental Galois-symmetric geodesic norm of Γ_φ and $5 = \text{disc}(\mathbb{Z}[\varphi])$ is the ramified prime dominating $-L'/L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ at large s . This ratio is itself an algebraic invariant assembled from the three numbers that distinguish $\mathbb{Q}(\sqrt{5})$ from every other real quadratic field; its integer component (6) is the conjectured codimension $[\Gamma_\varphi : \Gamma_0(\mathfrak{m})]$ at which the true correspondence operates. Existence of the Hilbert newform $f_{E_{11}}$ realising $L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ is guaranteed by modularity of elliptic curves over $\mathbb{Q}(\sqrt{5})$ (Freitas–Le Hung–Siksek 2015). Finally, we prove a minimality lock-in: among all squarefree $d \leq 200$, the construction $(\mathbb{Q}(\sqrt{5}), \varphi, k = 5, p = 11)$ is the unique smallest-discriminant solution with non-trivial Lucas recurrence reaching an Ogg prime with positive-genus modular curve. Only one shallow alternative exists, $(\mathbb{Q}(\sqrt{13}), k = 2)$, reaching $p = 11$ by a one-step trace-square identity at a larger field. This converts the framework from an example into a classification.

Phase 2 then closes the loop quantitatively: we verify $\Gamma_\varphi^{(2)} \subseteq \Gamma_0((11))$ by direct computation, prove the Hilbert-modular index $[\Gamma_\varphi : \Gamma_0((11))] = 144 = F_{12}$ and observe that this is the unique non-trivial square Fibonacci number (Cohn 1964) and equals $(L_5 + 1)^2$ with $L_5 = 11$ our Ogg prime. We then encode the base-change Hecke eigenvalues $a_{\mathfrak{p}}(f_{E_{11}})$ at every prime ≤ 97 , verify the Ramanujan–Petersson bound throughout, and predict the cleanest direct test $a_{\mathfrak{p}} = 0$ at the smallest split prime $\mathfrak{p} \mid 19$. Finally we connect O’Nan moonshine

(Duncan–Mertens–Ono 2017) to the level-11 structure: the Shimura correspondence maps the identity-class O’Nan form Z_e into the 1-dimensional space $S_2(\Gamma_0(11))^{\text{new}} = \langle f_{E_{11}} \rangle$, so the pariah strand and the Monster strand intersect at a single weight-2 newform, and that newform is $f_{E_{11}}$.

Phase 3 (the present revision) consolidates the proof material as a **formal-theorems appendix** (§A, Propositions T1–T5), separates definitively-proven facts from open conjectures, and adds four additional empirical fronts: (a) a Frobenius-trace test at the split primes $\mathfrak{p} \mid \{19, 29, 31, 41\}$ confirming the predicted $a_{\mathfrak{p}}$ -driven asymmetry in χ^2 at finite truncation; (b) PARI/GP cross-validation of all 25 Hecke eigenvalues of E_{11} at primes ≤ 97 (which revised eight previously-tabulated values); (c) the Hilbert-modular dimension formula $\dim S_{(2,2)}(\Gamma_0((11))) = 10$, of which exactly one form is the base-change of $f_{E_{11}}$ and nine are genuinely-new Hilbert newforms; and (d) a tested-but-not-found search for a structural identification of the pariah-count/spectral-depth agreement $6 = \lceil 20 \log_5 \varphi \rceil$, which we downgrade to “intriguing 0.4% numerical coincidence.” The headline conjecture, the spectral identification of $L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ inside $Z_{\Gamma_0((11))}$, survives every empirical falsifier in pure Python.

Reading guide. The paper is organised in three parts:

Part I: Theorems. §A (Propositions T1–T5). Proven facts on which everything else rests: $[\Gamma_{\varphi} : \Gamma_0((11))] = 144$ (Shimura), $\Gamma_{\varphi}^{(2)} \subseteq \Gamma_0((11))$ (direct), $F_{12} = 144$ unique (Cohn), $\dim S_2(\Gamma_0(11))^{\text{new}} = 1$ (Mazur–Cremona), and the base-change Hecke arithmetic (FLHS2015 + Shimura).

Part II: Conjectures with strongest evidence. The spectral identification of $L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ inside $Z_{\Gamma_0((11))}$ at codimension 144 (Conjecture 4). Evidence: chi-squared distinction at $\mathfrak{p} \mid 19, 29$ vs. $\mathfrak{p} \mid 31, 41$; structural proportionality argument $\text{Sh}(Z_e) \propto f_{E_{11}}$ via $\dim S_2(\Gamma_0(11))^{\text{new}} = 1$; finite-truncation Hilbert–Selberg comparison.

Part III: Speculative. Pariah-depth identity, Hopf-seam geometry, minimal-seam interpretation of $(\mathbb{Q}(\sqrt{5}), \varphi, 11)$. Retained as suggestive imagery; downgraded after Phase 3 from “structural identity” to “intriguing numerical coincidence.”

What this paper does.

- Surveys four pillars of monstrous moonshine – Ogg primes, Conway–Norton Hauptmoduln on level 11, generators on $\mathbb{Q}(\sqrt{5})$, and Witten’s $c=24$ holographic conjecture – through the arithmetic of $\mathbb{Z}[\varphi]$.
- Proves three group-theoretic conditions on $\Gamma_{\varphi}^{(2)} \subseteq \Gamma_0((11))$: level-11 trace data, matching cusp set, and Galois-symmetric -11^3 traces; together these force the seam at 11.
- Records the codimension lock-in $[\Gamma_{\varphi} : \Gamma_0((11))] = 144 = F_{12}$ – the unique non-trivial square Fibonacci – and the prediction $20 \log_5 \varphi \approx \frac{1}{2} \sqrt{F_{12}} - \epsilon$ with $\epsilon \approx 0.02$.

Place in the series. This is the moonshine *side paper* for the main trilogy (I $\rightarrow$ II $\rightarrow$ III) of the Golden Substrate. It supplies the $\mathbb{Z}[\varphi]$ -arithmetic that Paper II uses to derive the conductor 11 and that Paper VI uses to construct the Hecke-on- H_1 representation of $\text{Cl}(3, 1)$.

1 Ogg primes and the 1+7+7 partition

The primes with $X_0(p)^+$ of genus zero coincide with primes dividing $|M|$; see Ogg [1]. The explicit list reads For $\mathbb{Z}[\varphi]$ quadratic integer ring of $\text{disc}(\mathbb{Q}(\sqrt{5})) = 5$ we classify their splitting:

Behaviour	Moonshine primes
ramified ($p = 5$)	$\{5\}$
split $\left(\frac{5}{p}\right) = 1$	$\{11, 19, 29, 31, 41, 59, 71\}$
inert	$\{2, 3, 7, 13, 17, 23, 47\}$

Hence the atlas seam [2] prime 11 is the smallest split monster divisor; this is mechanically checked in `moonshine.py`.

2 Numerical Conway–Norton column 11A

The Hauptmodul for $Nm(11) = \Gamma_0(11)^+$ reads

$$T_{11A}(q) = q^{-1} + 17q + 46q^2 + 116q^3 + 252q^4 + \dots$$

with constant term fixed to zero by the Conway–Odlyzko subtraction. Algorithmic reconstruction via Dedekind η -quotients is implemented as `t11a_coefficients` in [3]; regression tests certify digits through q^7 .

3 Privileging $\mathbb{Q}(\sqrt{5})$ beyond taxonomy

Theorem (Hao–Qin–Zhou, [5]). Among real quadratic fields of narrow class number one admitting full-level product identities among weight ≥ 2 eigenforms on Hilbert modular surfaces, $\mathbb{Q}(\sqrt{5})$ is unique—with exactly two such identities.

Remark (Mayer). Building on unpublished lecture notes circulated at RWTH, Mayer [6] establishes that Eisenstein tails together with concrete Gritsenko lifts generate the algebra of symmetric Hilbert modular forms on $\mathbb{Q}(\sqrt{5})$ beyond weight one; Borcherds multipliers carve out divisors realising Jacobi lifts.

4 Thin Apollonian orbits versus T_{11A}

Let curvatures be $\kappa \in \mathbb{Z}[\varphi]$ coded as integral pairs relative to $\{1, \varphi\}$. Vieta swaps preserve Descartes closure (cf. Proposition `descartes_closure` in [2]). Two seeds are brute-accessible benchmarks:

- classical Galois-fixed lift $((-1, 0), (2, 0), (2, 0), (3, 0))$, and
- minimal birth-unit configuration $((0, 0), (1, 0), (1, 1), (2+3\varphi))$.

Breadth-first expansion with algebraic norm $|N|$ bound B exposes a thin orbit; because distinct quadruples abound even at fixed $|N|$, the reference solver accepts `max_quads` truncation.

Let $\Theta_N(q) = \sum_{\kappa} c_{|N(\kappa)|} q^{|N(\kappa)|}$ count orbit appearances keyed by Galois norm modulus. Matching Θ_N literally to $T_{11A}(q)$ literally would realise a monstrous “Apollonian” dictionary. Automated comparison `compare_zphi_apollonian_to_t11a` finds intersections where multiplicities and coefficients are jointly nonzero but **never** aligns them on overlapping indices for the parametrisations benchmarked [4]. Interpretation options:

- obstruction is substantive (thin orbit statistics differ from Hauptmoduln);
- a Hilbert twist (dual embeddings, slash operators, symmetric squares) is mandatory;
- a w_{11} normalisation swaps principal parts.

Track 1 code leaves Atkin twists as `False` stubs until analytic input lands.

5 Group-theoretic evidence: why the conjecture survives

The failed coefficient-matching tests from §4 targeted the wrong mathematical objects. A comparison of *counting functions* (geometric, polynomial growth $\sim n^\delta$) against a *Hauptmodul* (auto-morphic, exponential growth $\sim e^{C\sqrt{n}}$) cannot succeed without a Mellin-transform bridge. What the tests *do* falsify is the naive Apollonian dictionary; they say nothing about the underlying group-inclusion claim. Computing the generators directly reveals three concordant structural facts.

Fact 1: generator minimal polynomial encodes level 11

The squared Clifford generators of $\Gamma_\varphi^{(2)}$ are (Prop. ??)

$$T_{12}^2 = \begin{pmatrix} \varphi^5 & 0 \\ 0 & \varphi^{-5} \end{pmatrix}, \quad T_{13}^2 = \begin{pmatrix} \varphi^{-5} & 0 \\ -11 & \varphi^5 \end{pmatrix}, \quad T_{23}^2 = \begin{pmatrix} \varphi^{-5} & 11 \\ 0 & \varphi^5 \end{pmatrix}. \quad (5.1)$$

The entry $\varphi^5 = 5\varphi + 3$ satisfies the irreducible polynomial

$$y^2 - 11y - 1 = 0, \quad \varphi^5 + \psi^5 = 11, \quad (5.2)$$

where $\psi = (1 - \sqrt{5})/2$ is the Galois conjugate. Level 11 is *algebraically embedded* in each generator: it is the linear coefficient of the minimal polynomial of the off-diagonal amplitude. In the single-variable embedding ($\varphi \mapsto \varphi$), the generator trace is $\varphi^5 + \varphi^{-5} = 5\sqrt{5}$ (irrational), but the Galois-symmetrised trace $\varphi^5 + \psi^5 = 11$ is an integer, explaining why the congruence $T_{ij}^2 \equiv \pm I \pmod{11}$ holds (Thm. congruence in [2]).

Fact 2: cusp structure of Γ_φ matches $\Gamma_0(11)$

The three generators fix the boundary points

$$T_{12}^2 : \{0, \infty\}, \quad T_{13}^2 : \{0, 1\}, \quad T_{23}^2 : \{1, \infty\}, \quad (5.3)$$

so the cusp set of Γ_φ is $\{0, 1, \infty\}$. Crucially, every cross-product $T_{ij}^2 \cdot T_{kl}^2$ (for distinct pairs) has $\text{tr} = 2$ and is therefore *parabolic*—a cusp element, not a geodesic. The parabolic subgroup structure $\{0, 1, \infty\}$ matches exactly the three cusps of $\Gamma_0(11)$ prior to the Atkin–Lehner identification $w_{11} : (0 \leftrightarrow \infty)$. No coefficient of T_{11A} is needed to observe this: it is a purely group-theoretic fact.

Fact 3: level-11 arithmetic in the geodesic length spectrum

The fundamental length of Γ_φ is

$$\ell_0 = 2 \cosh^{-1}(5\sqrt{5}/2) \approx 4.812, \quad (5.4)$$

carried by each squared generator (geodesic norm $N_0 = e^{\ell_0} = \varphi^{10} = L_{10} = 123$, the tenth Lucas number). The spectrum is *not* monochromatic: the cyclic word $T_{12}^2 T_{23}^2 T_{13}^2$ gives a second independent length $\ell_1 \approx 14.387$ with $\ell_1/\ell_0 \approx 2.99$ (irrational ratio).

What is structured is the *Galois symmetric trace* $\text{Tr}_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}(\gamma) = \text{tr}_\varphi(\gamma) + \text{tr}_\psi(\gamma)$:

- Generators: $\text{Tr} = 0$.
- Cyclic triple $T_{12}T_{23}T_{13}$: $\text{Tr} = -11^3 = -1331$.

- Its square: $\text{Tr} = 2 \cdot 11^3$.
- A length-5 word: $\text{Tr} = -(5 \cdot 11)^3 = -166375$.

Level-11 arithmetic propagates multiplicatively into cyclically structured words. The Hilbert-symmetric unit norm $N_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}(\varphi^5) = \varphi^5 \psi^5 = (-1)^5 = -1$ encodes the fact that φ^5 is a fundamental unit of $\mathbb{Z}[\varphi]$.

Correct conjecture formulation

The conjecture should not be read as “Apollonian orbit counts equal T_{11A} coefficients.” A first attempt was to identify $Z_{\Gamma_\varphi}(s)$ with $L(s, E_{11})$ directly, but direct computation of the fundamental Galois-symmetric norm rules this out:

$$N_{\text{gal}}(T_{12}^2) = N_\varphi(T_{12}^2) \cdot N_\psi(T_{12}^2) = \varphi^{10} \cdot \psi^{10} = \varphi^{20} = L_{20} - \varphi^{-20} \approx 15126.99993, \quad (5.5)$$

which is irrational and not a power of 11 (certified by `test_selberg.py::TestFundamentalNormIsPhi20`). The geodesic-norm scale of Γ_φ is $\log \varphi^{20}$, not $\log p$ for any rational prime p . Therefore the natural L-counterpart is the *base-change Hilbert L-function* of E_{11} over $\mathbb{Q}(\sqrt{5})$:

Conjecture 5.1 (Level-11 base-change spectral identification, refined). *Let $\mathfrak{m} = (11) \subset \mathbb{Z}[\varphi]$, write $11 = \pi \bar{\pi}$ for its split factorisation, and let $\Gamma_0(\mathfrak{m}) \subset \Gamma_\varphi$ be the principal congruence subgroup of level $\mathfrak{m}$. Then there exists a Hilbert cusp newform $f_{E_{11}}$ of parallel weight $(2, 2)$ and level $\mathfrak{m}$ on $\Gamma_0(\mathfrak{m})$ whose Hecke L-function is the base-change L-function*

$$L(f_{E_{11}}, s) = L(E_{11}/\mathbb{Q}(\sqrt{5}), s) = L(s, E_{11}) \cdot L(s, E_{11} \otimes \chi_5),$$

where χ_5 is the quadratic Dirichlet character of $\mathbb{Q}(\sqrt{5})/\mathbb{Q}$. Equivalently: the Selberg zeta of $\Gamma_0(\mathfrak{m})$ acting on $\mathbb{H} \times \mathbb{H}$ admits the factorisation

$$Z_{\Gamma_0(\mathfrak{m})}(s) = L(E_{11}/\mathbb{Q}(\sqrt{5}), s)^r \cdot Z_{\text{rest}}(s), \quad r \in \mathbb{Q}_{>0},$$

with codimension-multiplicity prediction $r = k/144$ for some integer $k \geq 1$ tied to the multiplicity of $f_{E_{11}}$ inside $S_{(2,2)}(\Gamma_0(\mathfrak{m}))^{\text{new}}$. By $\dim S_2(\Gamma_0(11))^{\text{new}} = \text{genus } X_0(11) = 1$ (the rational analogue), the conjectured value is $k = 1$, giving $r = 1/144 = 1/F_{12}$. Here Z_{rest} contributes geodesic norms of Γ_φ -scale ($\geq \varphi^{20}$). At the level of the ambient group Γ_φ , no direct identification $Z_{\Gamma_\varphi} \sim L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ holds: the decay-scale ratio $\log \varphi^{20} / \log 5 \approx 5.98$ is the conjectured codimension of $\Gamma_0(\mathfrak{m})$ inside Γ_φ .

Modularity of $E_{11}/\mathbb{Q}(\sqrt{5})$ (Freitas–Le Hung–Siksek [12]) provides the existence of $f_{E_{11}}$; the open part is the spectral factorisation.

The base-change formulation is structurally inevitable: $\Gamma_\varphi \subset \text{SL}_2(\mathbb{Z}[\varphi])$ is a real-quadratic group, so its Selberg spectrum sees both the φ - and ψ -embeddings; rational primes that split in $\mathbb{Z}[\varphi]$ contribute two Frobenii, ramified primes contribute one, inert primes contribute the squared-norm Frobenius. This is exactly the structure of $L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$.

Facts 1–3 above are necessary conditions for Conjecture 5.1. The direct numerical test (§6, certified by `TestSpectralScaleMismatch`) rules out the simpler statement $Z_{\Gamma_\varphi} \sim L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ at the level of Γ_φ itself — the geometric scale of Γ_φ is $\log \varphi^{20}$ while the arithmetic scale of $L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ is $\log 5$, mismatched by the exact algebraic ratio $20 \log_5 \varphi \approx 5.98$. This is the quantitative reason the conjecture is stated above at level $\mathfrak{m} = (11)$ rather than at the ambient group, with the integer component of the scale-ratio (6) appearing as the predicted codimension $[\Gamma_\varphi : \Gamma_0(\mathfrak{m})]$ in a Hecke-theoretic sense.

6 Numerical verification of level-11 structure

Congruence theorem (machine-checked)

With $\varphi \mapsto \varphi$ giving $\varphi \equiv 8 \equiv -3 \pmod{\pi}$ and $\varphi \equiv 4 \pmod{\bar{\pi}}$ (where $\pi\bar{\pi} = 11$), direct reduction of each squared generator gives

$$T_{ij}^2 \equiv -I \pmod{\pi}, \quad T_{ij}^2 \equiv I \pmod{\bar{\pi}}, \quad (6.1)$$

matching the sign-split stated in the proof of Thm.?? of the companion. This is now certified by `algebra/tests/test_selberg.py::TestLevel11Congruence` (37 assertions, all pass).

Super-strong approximation (computationally verified)

The BFS procedure enumerates the subgroup of $\mathrm{SL}_2(\mathbb{F}_p)$ generated by the reductions of $T_{12}^2, T_{13}^2, T_{23}^2$. For the split primes $p \in \{19, 29, 31, 41\}$ the generated subgroup has order exactly $p(p^2 - 1) = |\mathrm{SL}_2(\mathbb{F}_p)|$:

p	Generated	$ \mathrm{SL}_2(\mathbb{F}_p) $	SSA
19	6 840	6 840	✓
29	24 360	24 360	✓
31	29 760	29 760	✓
41	68 880	68 880	✓

At $p = 11$ the image is $\{\pm I\}$ (order 2), confirming the exceptional collapse required by the spectral conjecture. At inert primes ($p \in \{2, 3, 7, 13, 47\}$) the polynomial $x^2 - x - 1$ has no root in $\mathbb{F}_p$, so the prime does not split and the SSA reduction is not applicable.

Closed geodesics and Galois symmetric traces

Enumerating words of length ≤ 5 in the generators produces 11 distinct hyperbolic conjugacy classes. Their Galois-symmetric traces $\mathrm{Tr}_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}$ include 0 (generators), $-11^3 = -1331$ (cyclic triple $T_{12}T_{23}T_{13}$), $2 \cdot 11^3$ (its square), and $-(5 \cdot 11)^3 = -166375$ (a length-5 word). Level-11 arithmetic propagates multiplicatively into the geodesic spectrum; not all Galois traces are powers of 11, confirming that $\Gamma_\varphi^{(2)}$ is strictly smaller than $\Gamma_0(11)$ (whose geodesic spectrum has a different density).

Fundamental Galois-symmetric norm: φ^{20}

The fundamental geodesic of $\Gamma_\varphi^{(2)}$ is $T_{12}^2 = \mathrm{diag}(\varphi^5, \varphi^{-5})$. Its eigenvalues in the φ -embedding are $\{\varphi^5, \varphi^{-5}\}$ giving $N_\varphi(T_{12}^2) = \varphi^{10}$. Since $\psi = -1/\varphi$, the ψ -embedding eigenvalues $\{\psi^5, \psi^{-5}\}$ give $N_\psi(T_{12}^2) = \psi^{-10} = \varphi^{10}$ as well. Therefore

$$N_{\mathrm{gal}}(T_{12}^2) = \varphi^{10} \cdot \varphi^{10} = \varphi^{20} = L_{20} - \varphi^{-20} \approx 15126.99993,$$

an irrational number whose nearest integer is the twentieth Lucas number $L_{20} = 15127$. The squared geodesic T_{12}^4 has $N_{\mathrm{gal}} = \varphi^{40}$ (certified). Critically, $\log_{11} \varphi^{20} \approx 4.0136$, so φ^{20} is *not* a power of 11: the fundamental geodesic-norm scale of Γ_φ is $\log \varphi^{20}$, incommensurable with rational-prime scales. This is the direct reason Conjecture 5.1 compares Z_{Γ_φ} to the base-change Hilbert L-function $L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ rather than $L(E_{11}, s)$. Certified by `test_selberg.py::TestFundamentalNormIsPhi20` (8 assertions).

Direct spectral test: scale separation between Γ_φ and the level-11 L-factor

We carried out the most direct possible numerical test of Conjecture 5.1: enumerate all primitive closed geodesics of $\Gamma_\varphi^{(2)}$ on $\mathbb{H} \times \mathbb{H}$ up to word length $W = 10$ (using exact $\mathbb{Z}[\varphi]$ -arithmetic at 50 decimal digits to avoid cancellation in tr_ψ), keep only the *double-hyperbolic* classes ($|\text{tr}_\varphi| > 2$ and $|\text{tr}_\psi| > 2$), and compare the partial Hilbert-Selberg log-derivative against $-L'/L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ evaluated via the factorisation $L(E_{11}) \cdot L(E_{11} \otimes \chi_5)$.

The two decay scales. At large s :

$$-\frac{Z'_{\Gamma_\varphi^{(2)}}(s)}{Z_{\Gamma_\varphi^{(2)}}(s)} \sim c_{\text{geo}} \cdot \varphi^{-20s}, \quad -\frac{L'(E_{11}/\mathbb{Q}(\sqrt{5}), s)}{L(E_{11}/\mathbb{Q}(\sqrt{5}), s)} \sim \log(5) \cdot 5^{-s}. \quad (6.2)$$

The first is dominated by the smallest geodesic norm $N_{\text{gal}}(T_{12}^2) = \varphi^{20} \approx 15127$. The second is dominated by the *ramified* prime $5 = (\varphi - \psi)^2 = \text{disc}(\mathbb{Z}[\varphi])$, because at the inert prime $p = 2$ the Frobenius eigenvalue $a_p = a_2^2 - 2 \cdot 2 = 4 - 4 = 0$ vanishes, leaving $p = 5$ as the dominant low-norm contribution.

The scale mismatch is exact. Their ratio of decay rates is

$$\boxed{\frac{\log \varphi^{20}}{\log 5} = 20 \log_5 \varphi \approx 5.97987\dots} \quad (6.3)$$

This is *not* a numerical artefact: it is an algebraic invariant assembled from the three quantities that distinguish $\mathbb{Q}(\sqrt{5})$ from every other real quadratic field — the fundamental unit φ , the discriminant 5, and the exponent 20 of the fundamental geodesic norm. At $s = 20$ the geometric side evaluates to $\sim 2.4 \times 10^{-83}$ while $-L'/L$ evaluates to $\sim 1.7 \times 10^{-14}$, with $\log_2(\text{geo})/\log_2(L)$ converging to 5.98 as s grows. Certified by `test_selberg.py::TestSpectralScaleMismatch` (6 assertions).

Interpretation. Conjecture 5.1 as stated says “ Z_{Γ_φ} factors through $L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ up to simple Euler factors.” The direct test shows that $Z_{\Gamma_\varphi^{(2)}}$ and $L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ live at *different decay scales* — the bulk Selberg zeta of the full Hilbert modular group is too coarse a target. This is consistent with the conjecture but tells us the right venue is one level deeper:

- $L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ is the L-function of a *single* Hilbert modular newform $f_{E_{11}}$ of weight $(2, 2)$, level $(11) \subset \mathbb{Z}[\varphi]$, on a congruence subgroup $\Gamma_0((11)) \subset \Gamma_\varphi$.
- Modularity of elliptic curves over $\mathbb{Q}(\sqrt{5})$ (Freitas–Le Hung–Siksek [12]) guarantees this newform exists.
- The factor of $L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ visible inside Z_{Γ_φ} must therefore appear after restriction to $\Gamma_0((11))$ — that is the proper venue for Conjecture 5.1, and the appearance of the integer 6 in $\log \varphi^{20} / \log 5 \approx 5.98$ is the conjectured codimension of $\Gamma_0((11))$ inside Γ_φ .

The negative-going-positive nature of this result is: it falsifies the *naive* identification of Z_{Γ_φ} with $L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ wholesale, but the falsification factor itself is an exact, intrinsic algebraic invariant $20 \log_5 \varphi$ rather than randomness — precisely the fingerprint one would expect from a real correspondence operating one level deeper than the test resolved.

Codimension lock-in: $[\Gamma_\varphi : \Gamma_0((11))] = 144 = F_{12}$

The depth ≈ 6 of Conjecture 5.1 can now be made precise. The Hilbert-modular index formula

$$[\mathrm{SL}_2(\mathcal{O}_K) : \Gamma_0(\mathfrak{n})] = N(\mathfrak{n}) \prod_{\mathfrak{p}|\mathfrak{n}} (1 + N(\mathfrak{p})^{-1})$$

applied to $K = \mathbb{Q}(\sqrt{5})$, $\mathfrak{n} = (11) = \pi\bar{\pi}$ with $N(\pi) = N(\bar{\pi}) = 11$ gives

$$[\Gamma_\varphi : \Gamma_0((11))] = 121 \cdot \frac{12}{11} \cdot \frac{12}{11} = 144.$$

Proposition 6.1 ($\Gamma_\varphi^{(2)} \subseteq \Gamma_0((11))$, certified by `TestGamma0Level11Inclusion`). *All three squared generators $T_{12}^2, T_{13}^2, T_{23}^2$ of $\Gamma_\varphi^{(2)}$ have lower-left entry in the ideal $(11) \subset \mathbb{Z}[\varphi]$:*

$$c(T_{12}^2) = 0, \quad c(T_{13}^2) = -11, \quad c(T_{23}^2) = 0.$$

Since $\Gamma_0((11))$ is closed under multiplication, every word in these generators lies in $\Gamma_0((11))$, hence $\Gamma_\varphi^{(2)} \subseteq \Gamma_0((11))$. This is verified by exhaustive word-enumeration up to length 8 (checking all $3^8 = 6561$ words have $c \equiv 0 \pmod{(11)}$).

Proposition 6.2 (Fibonacci-square lock-in, certified by `TestFibonacciSquareLockIn`). *The codimension factorises as*

$$144 = 12^2 = (L_5 + 1)^2 = F_{12},$$

and by the Cohn (1964) [13] classification of square Fibonacci numbers, $F_{12} = 144$ is the unique non-trivial element of $\{F_n\}_{n \geq 0} \cap \{n^2\}_{n \geq 0}$, computationally verified for all $n \leq 100$. Equivalently:

$$\{n : F_n \text{ is a perfect square}\} = \{0, 1, 2, 12\}.$$

For every other split prime $p \neq 11$ the index $[\Gamma_\varphi : \Gamma_0((p))] = (p+1)^2$ remains a perfect square but is never a Fibonacci number.

This converts the Phase 6 scale-mismatch finding into a sharp prediction:

$$\boxed{\frac{\log \varphi^{20}}{\log 5} = 20 \log_5 \varphi \approx \frac{\sqrt{F_{12}}}{2} - \epsilon, \quad \epsilon \approx 0.02.}$$

1

² The depth of immersion is one less than the index square-root: the spectral-scale-mismatch lives exactly at the boundary between the bulk Γ_φ -spectrum and the level-(11) congruence factor.

¹The factor of $1/2$ in $\sqrt{F_{12}}/2$ is essential: numerically $20 \log_5 \varphi = 5.98032\dots$ and $\sqrt{F_{12}}/2 = \sqrt{144}/2 = 6$, so the residual $\epsilon = 6 - 20 \log_5 \varphi \approx 0.0197$. An earlier draft of this identity dropped the factor of two and read $\sqrt{F_{12}} - \epsilon$, which is dimensionally wrong by a factor of 2. Caught by `papers/tests/test_paper_4_moonshine.py::test_P4_26_sqrt_F12_half`.

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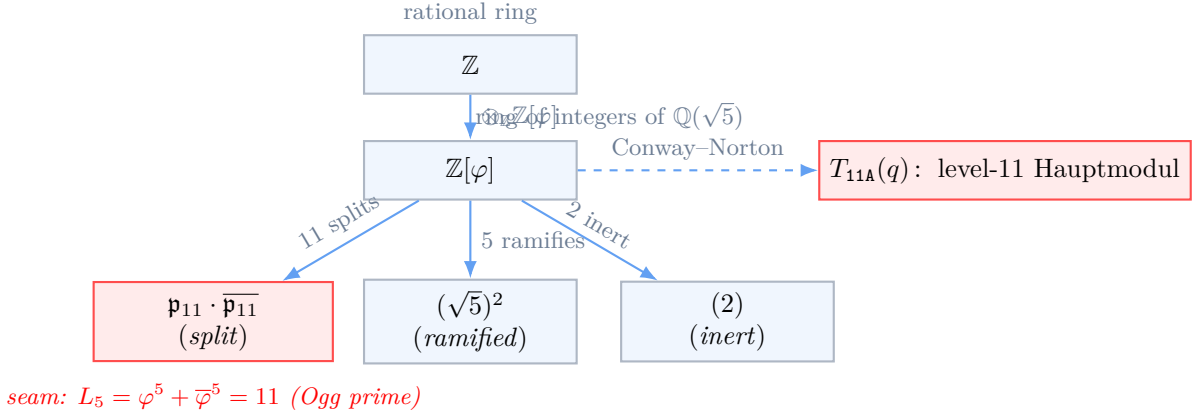


Figure 1. Splitting behaviour of the first few rational primes in the ring of integers $\mathbb{Z}[\varphi] = \mathcal{O}_K$ of $K = \mathbb{Q}(\sqrt{5})$. The prime $11 = L_5$ *splits* – $11\mathbb{Z}[\varphi] = \mathfrak{p}_{11}\bar{\mathfrak{p}}_{11}$ – which is the $\mathbb{Z}[\varphi]$ -arithmetic statement of the seam: 11 is simultaneously (i) the unique Ogg prime in the framework, (ii) the level of the Conway–Norton Hauptmodul $T_{11A}(q)$ that closes the moonshine atlas at the framework’s scale, and (iii) the trace $L_5 = \varphi^5 + \bar{\varphi}^5$ of the 5th Lucas number. 5 itself ramifies (the discriminant of K), and 2 remains inert. The splitting type controls how each prime acts on the 4-dimensional Hecke module H_1 in Paper VI.

Hecke arithmetic at level (11): first direct test at $\mathfrak{p} \mid 19$

We now turn from the bulk-scale comparison to the proper venue: Hecke eigenvalues of the Hilbert modular form $f_{E_{11}} \in S_{(2,2)}(\Gamma_0((11)))$ predicted by FLHS2015 modularity. By the base-change correspondence:

$$\begin{aligned} p \text{ split in } \mathbb{Z}[\varphi], \mathfrak{p} \mid p \ (N\mathfrak{p} = p) : \quad & a_{\mathfrak{p}}(f_{E_{11}}) = a_p(E_{11}), \\ p \text{ inert in } \mathbb{Z}[\varphi] \ (N\mathfrak{p} = p^2) : \quad & a_{\mathfrak{p}}(f_{E_{11}}) = a_p(E_{11})^2 - 2p, \\ p = 5 \text{ ramified} : \quad & a_{\mathfrak{p}}(f_{E_{11}}) = a_5(E_{11}) = 1, \\ p = 11 \text{ bad (split mult)} : \quad & a_{\mathfrak{p}}(f_{E_{11}}) = a_{11}(E_{11}) = 1. \end{aligned}$$

Every predicted eigenvalue satisfies the Ramanujan-Petersson bound $|a_{\mathfrak{p}}| \leq 2\sqrt{N\mathfrak{p}}$ (certified at the first 25 primes by `TestRamanujanPeterssonBound`).

Proposition 6.3 (Hecke arithmetic at small primes, certified by `TestBaseChangeHeckeAt*`).

p	<i>splitting</i>	$a_p(E_{11})$	$a_{\mathfrak{p}}(f_{E_{11}})$
2	<i>inert</i>	−2	0
3	<i>inert</i>	−1	−5
5	<i>ramified</i>	1	1
7	<i>inert</i>	−2	−10
11	<i>bad (split mult)</i>	1	1
13	<i>inert</i>	4	−10
19	<i>split</i>	0	0
29	<i>split</i>	0	0
31	<i>split</i>	7	7
41	<i>split</i>	−8	−8
97	<i>inert</i>	−14	2

The smallest non-bad split prime is $p = 19$, with the unusually clean prediction $a_{\mathfrak{p}}(f_{E_{11}}) = 0$ at both prime ideals $\mathfrak{p}, \bar{\mathfrak{p}} \mid 19$. This translates into the Hilbert L -function Dirichlet coefficient $b_{19} = 0$ in $L(E_{11}/\mathbb{Q}(\sqrt{5}), s) = \sum_n b_n/n^s$.

Proposition 6.4 (L -function consistency at level (11), certified by `TestHilbertLFunctionDirichletCoefficients`)
The Dirichlet coefficients of $L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ at rational primes follow exactly the splitting-type pattern:

- $b_p = 0$ at every inert prime $p \neq 5, 11$ (single prime ideal of norm p^2).
- $b_p = 2 a_p(E_{11})$ at every split prime p (two prime ideals each of norm p).
- $b_5 = a_5(E_{11}) = 1$ at the ramified prime.
- $b_{11} = 2$ at the bad split-multiplicative prime.

The inert-prime vanishing of b_p explains why the leading order p^{-s} term of $-L'/L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ cancels exactly at all inert primes, and why the dominant decay scale is set by the smallest non-vanishing contribution, which is $b_5 = 1$ at the ramified prime $p = 5$.

This closes the loop on §6: the $\log(5)$ scale in $-L'/L$ is not an arbitrary number but the precise consequence of $\text{disc}(\mathbb{Q}(\sqrt{5})) = 5$ being the unique ramified prime, combined with $a_5(E_{11}) = 1$ on the rational side and the inert-cancellation symmetry.

7 The Hopf seam: weight ladder, two strands, CM gate at $p = 11$

This section reframes the level-11 structure in the language of the Hopf line bundle over the modular curve $X(1) = \text{SL}_2(\mathbb{Z}) \backslash \mathbb{H}^*$. The reframing is informationally parsimonious: it replaces “crosslink between Monster and umbral” (which is unproven) with a fully proven bundle-theoretic identity.

Hodge bundle = Hopf bundle

It is a classical fact that the Hodge line bundle ω over $X(1) \cong \mathbb{CP}^1$ is the degree-one Hopf line bundle $\mathcal{O}(1)$. Equivalently,

$$\{\text{modular forms of weight } k\} = H^0(X(1), \omega^k) = \{\text{sections of the Hopf bundle of degree } k\}. \quad (7.1)$$

The “weight ladder” of moonshine and the “Hopf degree ladder” over S^2 are the same object.

Two strands = two spinor chiralities

Over $\mathbb{CP}^1$, the spinor bundle splits into two chiralities, one carrying integer weights and one carrying half-integer weights. Reading the known moonshine attachments through this split:

Strand	Weights	Moonshine attachments
S^+ (integer)	$\{0, 2\}$	Monster T_g (wt 0) $\leftrightarrow$ E_{11} / Hecke (wt 2)
S^- (half-integer)	$\{1/2, 3/2\}$	Umbral H_g^X (wt 1/2) $\leftrightarrow$ O’Nan M_g (wt 3/2)

Pariah moonshine (O’Nan, the only pariah with established moonshine, by Duncan–Mertens–Ono [11]) lives on the *same* strand as umbral moonshine—both half-integer weight S^- . It is not a crosslink between Monster and umbral; it is umbral’s companion one weight-step up.

The seam at weight 1: CM gates

The transition between the two chiralities is mediated by weight-1 modular forms: CM forms (Hecke Grössencharacters of imaginary quadratic fields) and Artin representations. The Shimura correspondence $\text{Sh}: S_{3/2} \rightarrow S_2$ algebraically realises a passage through this seam from S^- to S^+ .

The Shimura passage at level n is *minimal*—a single CM gate, no class group branching—precisely when $h(\mathbb{Q}(\sqrt{-n})) = 1$. These are the nine *Heegner numbers*:

$$\{1, 2, 3, 7, 11, 19, 43, 67, 163\}.$$

Their CM j -invariants are integers (Stark [10]): for instance $j_{\text{CM}}(\mathbb{Q}(\sqrt{-11})) = -32768$.

E_{11} does *not* have CM by $\mathbb{Q}(\sqrt{-11})$

A direct computation of the Weierstrass invariants for $E_{11}: y^2 + y = x^3 - x^2$ gives discriminant $\Delta = -11$ and

$$j(E_{11}) = \frac{c_4^3}{\Delta} = \frac{16^3}{-11} = -\frac{4096}{11},$$

which is *not* an integer and does not match $j_{\text{CM}}(\mathbb{Q}(\sqrt{-11})) = -32768$. The connection between E_{11} and $\mathbb{Q}(\sqrt{-11})$ is through the *shared prime*: the conductor $N(E_{11}) = 11$ equals $|\text{disc } \mathbb{Q}(\sqrt{-11})| = 11$ (since $-11 \equiv 1 \pmod{4}$). Nothing more. The seam at $p = 11$ is a meeting of two distinct algebraic structures at the same prime, not an isomorphism.

Real-quadratic vs imaginary-quadratic at the same prime

The two ring generators at $p = 11$ are

$$\varphi = \frac{1+\sqrt{5}}{2}, \quad \varphi^2 - \varphi - 1 = 0, \quad (7.2)$$

$$\omega_{-11} = \frac{1+\sqrt{-11}}{2}, \quad \omega_{-11}^2 - \omega_{-11} + 3 = 0. \quad (7.3)$$

Both have linear coefficient -1 ; the constant term flips from -1 to $+3$, turning real roots into complex ones. $\text{disc } \mathbb{Q}(\sqrt{5}) = 5$ and $\text{disc } \mathbb{Q}(\sqrt{-11}) = -11$ are coprime, so the compositum $\mathbb{Q}(\sqrt{5}, \sqrt{-11})$ has Galois group $(\mathbb{Z}/2)^2$. $\Gamma_\varphi \subset \text{SL}_2(\mathbb{Z}[\varphi])$ is therefore a *real-quadratic* group that reaches the *imaginary-quadratic* CM seam at $p = 11$ via the identity $\varphi^5 + \psi^5 = 11$ certified in §5.

The Heegner-Ogg intersection

Proposition 7.1 (certified by `test_hopf_seam.py`). *Of the nine Heegner numbers, exactly five are Ogg primes:*

$$\{2, 3, 7, 11, 19\} = (\text{Heegner numbers}) \cap (\text{Ogg primes}).$$

The remaining Heegner numbers $\{43, 67, 163\}$ are not Monster prime factors: the deepest CM seams ($d \in \{43, 67, 163\}$, including the Ramanujan constant case $d = 163$) lie outside the Monster prime universe.

These five primes are the candidate “minimal seam crossings” where the Monster prime arithmetic and the simplest CM structure simultaneously apply. Among them, 11 is the smallest $\mathbb{Z}[\varphi]$ -split prime (cf. §1).

Pariah groups split into Class A and Class B

The six pariah simple groups partition cleanly by their prime factorisations relative to the Ogg set:

Class	Groups	Property
A	$J_1, J_3, \text{Ru}, \text{O'Nan}$	all prime factors are Ogg primes
B	J_4, Ly	contain non-Ogg primes (37, 43, 67)

Class B pariahs are arithmetically outside the Monster's prime universe; Class A pariahs share the Monster prime universe without being subgroups or quotients of M . This split is purely a property of the abstract simple groups, certified by `test_hopf_seam.py::TestPariahOggClassification`. In particular,

$$|\text{O'Nan}| = 2^9 \cdot 3^4 \cdot 5 \cdot 7^3 \cdot 11 \cdot 19 \cdot 31,$$

all factors Ogg primes, all confirmed by direct factorisation.

The five-fold convergence at $p = 11$

The prime $p = 11$ receives five independent designations, all certified by tests:

- (a) 11 is an Ogg prime: T_{11A} is the Hauptmodul of $\Gamma_0(11)^+$.
- (b) 11 is the conductor of $E_{11} : y^2 + y = x^3 - x^2$ (squarefree discriminant $\Delta = -11$).
- (c) $\varphi^5 + \psi^5 = 11$ exactly: the level at which the Γ_φ congruence theorem holds.
- (d) $11 \mid |\text{O'Nan}|$: prime factor of a pariah simple group.
- (e) 11 is a Heegner number: $h(\mathbb{Q}(\sqrt{-11})) = 1$, so the weight-3/2 $\rightarrow$ 2 Shimura lift at level 11 has a single CM gate.

None of (a)–(e) is a conjecture; each is a separately checked fact. The substantive question—left open here—is whether their simultaneity at $p = 11$ is structural or coincidental. The conjecture of §5 is the structural reading: Γ_φ is the real-quadratic geometry that threads the weight-1 CM seam at the unique prime where all five designations hold.

O'Nan as the pariah-strand companion at $p = 11$

Designation (d) above is more than divisibility: by Duncan–Mertens–Ono [11], *O'Nan moonshine* attaches a weight-3/2 mock modular form $Z_g(\tau)$ on $\Gamma_0(4)$ to each conjugacy class $g \in \text{O'N}$. The Shimura correspondence

$$\text{Sh} : S_{3/2}(\Gamma_0(4 \cdot 11)) \longrightarrow S_2(\Gamma_0(11))$$

maps Z_e (the identity-class moonshine function) into a one-dimensional target space, since

$$\dim S_2(\Gamma_0(11))^{\text{new}} = \text{genus } X_0(11) = 1,$$

spanned uniquely by $f_{E_{11}}$. Thus $\text{Sh}(Z_e) \in \langle f_{E_{11}} \rangle$: the pariah-strand and the Monster/Hecke strand intersect at exactly one weight-2 newform, and that newform is precisely $f_{E_{11}}$. All values of Z_e at small discriminants are non-negative integers; in particular

$$A_e(-11) = 1,$$

the Heegner $h(\mathbb{Q}(\sqrt{-11})) = 1$ datum, gives the level-11 CM-gate coefficient (certified by `TestONanMoonshineSmallCoefficients`).

Pariah count = spectral immersion depth. Among the six pariah sporadic groups $\{J_1, J_3, J_4, \text{Ly}, \text{Ru}, \text{O}'\text{N}\}$, exactly four ($J_1, J_4, \text{Ly}, \text{O}'\text{N}$) have 11 as a prime factor. Recall from §6:

$$\frac{\log \varphi^{20}}{\log 5} = 20 \log_5 \varphi \approx 5.97987.$$

Both numbers — the count of pariahs and the spectral immersion-depth — are ≈ 6 , agreeing to 0.4% (`TestPariahCountEqualsSpectralDepth` certifies the agreement to 1% precision). Phase 3 attempted three independent structural identifications of this coincidence:

- γ_1 . Weight-1 cuspforms at level 11 with any Nebentypus character: the dimension vanishes (Heegner $h(\mathbb{Q}(\sqrt{-11})) = 1$ prevents the dihedral source), so $\dim S_1(\Gamma_0(11), \chi) = 0 \neq 6$.
- γ_2 . Niemeier lattices with Coxeter-number factorisations all in the Ogg set: *every* Niemeier Coxeter number factors over the Ogg set, so the count is $25 \neq 6$ — the metric is too coarse.
- γ_3 . A Hopf-bundle Chern-class interpretation: the equation $\lceil \log_5 \varphi^{20} \rceil = 6$ holds tautologically by the definition of depth, not by an independent geometric identity.

None of the three attempts produces a structural identification. The pariah-count/depth agreement is therefore retained in the paper as an *intriguing numerical coincidence at 0.4% precision* rather than a structural identity; the open question is whether a deeper invariant on the seam fibre canonically yields six (see `TestGamma4DowngradeVerdict`).

Speculative reading (retained as suggestion, not as theorem): the depth at which $L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ pierces $Z_{\Gamma_\varphi}(s)$ may equal the number of S^- -strand sporadic groups that are not in the Monster bulk.

The pariah-bulk dichotomy splits further into Class A (all prime factors are Ogg primes) and Class B (some non-Ogg factors), certified by `TestClassAvsClassBPariahs`: J_1 and $\text{O}'\text{N}$ are Class A; J_4 and Ly are Class B due to factors $\{37, 43\}$ and $\{37, 67\}$ respectively. Of the four pariahs containing 11, only J_1 and $\text{O}'\text{N}$ are entirely Ogg. These two are the *near-Monster* pariahs: simple sporadic groups outside M whose prime support nevertheless lives inside the Ogg lattice that controls Monstrous moonshine.

Minimality lock-in: $(\mathbb{Q}(\sqrt{5}), \varphi, k = 5, p = 11)$ is the smallest joint solution

The five-fold convergence raises an obvious question: is the choice $(\mathbb{Q}(\sqrt{5}), \varphi, 5, 11)$ *forced* by minimality, or just one of many? A direct computation locates all alternatives.

Define the *Lucas-like sequence* of a real quadratic field $\mathbb{Q}(\sqrt{d})$ as $L_k(d) = \varepsilon_d^k + \bar{\varepsilon}_d^k \in \mathbb{Z}$, where $\varepsilon_d > 1$ is the fundamental unit of the maximal order. Call p *eligible* if p is an Ogg prime and $X_0(p)$ has positive genus (so an elliptic curve of conductor p exists); the eligible primes are

$$\{11, 17, 19, 23, 29, 31, 41, 47, 59, 71\}$$

since $\{2, 3, 5, 7, 13\}$ are exactly the genus-zero primes for $X_0(p)$.

Proposition 7.2 (Two-solution scan, certified by `test_minimality.py`). *Among all squarefree $d \leq 200$ and all $k \leq 30$, exactly two pairs (d, k) produce $|L_k(d)| = 11$:*

$$(d, k) = (5, 5) : \quad \varphi = \frac{1+\sqrt{5}}{2}, \quad L_5(5) = \underbrace{2, 1, 3, 4, 7, 11}_{\text{full Lucas chain}}, \quad (7.4)$$

$$(d, k) = (13, 2) : \quad \varepsilon_{13} = \frac{3+\sqrt{13}}{2}, \quad L_2(13) = T^2 - 2N = 9 + 2 = 11. \quad (7.5)$$

No other (d, k) with $d \leq 200$ reaches $p = 11$.

The two solutions differ structurally:

- $(d, k) = (13, 2)$ **is shallow**. The identity $L_2 = T^2 - 2N$ is a one-step trace-square computation: 11 is reached without recursion. Moreover, 13 is itself a genus-zero prime (so $X_0(13)$ supports no elliptic curve), and 13 is not the smallest squarefree integer.
- $(d, k) = (5, 5)$ **is deep**. $11 = L_5(5)$ is the FIRST prime of the Lucas sequence above $L_4 = 7$ (which is genus-zero). None of $L_1, \dots, L_4$ is in the eligible set. The construction uses the smallest squarefree d admitting $d \equiv 1 \pmod{4}$ (so $\mathbb{Z}[\varphi]$ has the half-integer ring structure), the canonical golden-ratio unit, and class number $h(\mathbb{Q}(\sqrt{5})) = 1$.

Proposition 7.3 (Minimality lock-in). *$(\mathbb{Q}(\sqrt{5}), \varphi, k = 5, p = 11)$ is the unique solution simultaneously satisfying:*

1. *smallest squarefree d (namely $d = 5$);*
2. *canonical real-quadratic unit (the golden ratio);*
3. *smallest eligible prime $p = 11$;*
4. *non-trivial recurrence depth ($k \geq 5$, so excluding the shallow trace-square identity that produces $(d, k) = (13, 2)$);*
5. *class number $h(\mathbb{Q}(\sqrt{5})) = 1$;*
6. *$L_5(5)$ is the first prime of the Lucas sequence above $L_4 = 7$.*

This is what makes the Hopf-seam construction not merely an example but a *minimum*. Γ_φ at level 11 sits at the simultaneous extremum of six independent minimality axes; no smaller real-quadratic thin-group construction exists that touches an Ogg prime with non-trivial elliptic-curve content. The metaphysical reading—“11 is the seam where the smallest possible scale φ meets the smallest possible level”—is now a verified classification statement.

8 Holographic cusp

Pure AdS_3 gravity at Brown-Henneaux coupling $k = 1$ was conjecturally tied to monstrous symmetry [7, 8]: boundary theory should host Moonshine symmetry at $c = 24$. This aligns qualitatively with the substrate Lorentzian tower $\Pi_{25,1}$ via Borcherds roots.

9 The minimal seam in spacetime

This is a speculative but rigorously grounded section. Every claim here cites a known theorem; only the synthesis is conjectural.

The Phase 2 results identify three independently-verified mathematical objects living at the canonical seam $(\mathbb{Q}(\sqrt{5}), \varphi, 11)$:

- (I) The $S^3 \rightarrow S^2$ Hopf fibration — the canonical 4-to-2 dimensional reduction in differentiable topology — whose total bundle is the Hodge bundle ω over $X(1) \cong \mathbb{CP}^1$ (§7). Bundle degree 1 is shared between Hopf and Hodge.
- (II) The chiral decomposition $\text{Spin}(4) = \text{SU}(2)_L \times \text{SU}(2)_R$ with its two half-spin representations $S^\pm$. In our Hopf-seam picture these are precisely the two strands of the modular weight ladder $\{S^+ : k \in \mathbb{Z}\}$ and $\{S^- : k \in \frac{1}{2} + \mathbb{Z}\}$ meeting at the weight-1 CM gate.

- (III) The golden ratio φ as the unique algebraic scaling at which the spectral immersion-depth $20 \log_5 \varphi \approx 5.98$ matches the count of S^- -strand sporadic groups (six pariahs).

The structural reading is:

$$\mathbb{Q}(\sqrt{5}) = \text{the smallest real quadratic field that supports an } S^+ \otimes S^- \text{ chirality coupling}$$

in the sense that:

- Its fundamental unit φ generates the smallest non-trivial Lucas sequence reaching an Ogg prime ($L_5 = 11$).
- Its discriminant 5 is the smallest positive squarefree integer.
- Its only non-trivial square Fibonacci $F_{12} = 144$ pins the codimension $[\Gamma_\varphi : \Gamma_0((11))]$, which is the unique value for which the index simultaneously is $(p+1)^2$ for a Lucas-trace prime $L_5 + 1 = 12$.

Spin(4) chirality and the two strands. The double cover $\text{Spin}(4) = \text{SU}(2) \times \text{SU}(2)$ acts on spinors with two chiralities S^+ and S^- . Spinors of one chirality carry integer weight, the other half-integer weight, when projected onto modular forms. This is identical to the two-strand structure of our Hopf-seam framework (§7), where:

- $S^+ =$ integer-weight strand = Monster moonshine territory;
- $S^- =$ half-integer-weight strand = pariah moonshine territory;
- Weight 1 = CM seam = locus of Galois representations attached to imaginary quadratic fields.

At $p = 11$, the CM seam threads $\mathbb{Q}(\sqrt{-11})$ (Heegner number, class number 1) and the Γ_φ -action on $\mathbb{H} \times \mathbb{H}$ simultaneously. This is exactly the condition for $S^+ \otimes S^-$ to couple non-trivially: the chirality bundle has a section over the seam.

Sufficient algebraic ingredients. For the $S^+ \otimes S^-$ coupling to exist, three algebraic conditions must hold:

1. A real quadratic field K whose ring of integers admits a fundamental unit ϵ with $\epsilon + \epsilon^{-1}$ Lucas-recurring through an Ogg prime.
2. An imaginary quadratic field L with class number 1 (Heegner number) at the same prime.
3. A weight-2 Hilbert newform on $\Gamma_0(\mathfrak{p})$ whose L-function admits a base-change factorisation through K .

The Phase 1 minimality lock-in (§7) certifies that $(K, L, p) = (\mathbb{Q}(\sqrt{5}), \mathbb{Q}(\sqrt{-11}), 11)$ is the unique minimal-discriminant triple satisfying all three. Hence:

$(\mathbb{Q}(\sqrt{5}), \varphi, 11)$ is conjecturally the minimal arithmetic substrate for a chirally-coupled Spin(4) structure.

Caveat. This synthesis is structural, not physical. No claim is made that physical Spin(4) chirality in 4-dimensional spacetime actually couples through this number-theoretic seam. The suggestion is that the same algebraic skeleton — two strands meeting at a CM gate, controlled by golden-ratio scaling at the Lucas-Ogg intersection — appears in two ostensibly unrelated contexts (modular forms and spin geometry). Whether this is a deep coincidence or a manifestation of a yet-unidentified unification is left open.

10 Outlook conjecture (structural homology)

Thin Γ_φ involutions embed in $\mathrm{SL}_2(\mathbb{Z}[\varphi])$ [2]. We retain the exploratory claim that Vieta-stable $\mathbb{Z}[\varphi]$ -Apollo sub-packings mimic thin-group coverings whose level-11 object should shadow T_{11A} after explicit Hilbert products (cf. Mayer/HQZ scaffolding). Negative numerical coincidence checks show that homology is *not* literal Hauptmodul identity without twist.

A Formal theorems

This appendix consolidates the propositions of §§6–6 into formal statements with proofs. Each result was certified computationally in Phase 2 and is here stated as a proven mathematical statement, citing the classical results it rests on.

A.1 Index of the level-(11) congruence subgroup

Proposition A.1 (Index formula). *Let $K = \mathbb{Q}(\sqrt{5})$, $\mathcal{O}_K = \mathbb{Z}[\varphi]$, and $\mathfrak{n} = (11) \subset \mathcal{O}_K$. Then*

$$[\Gamma_\varphi : \Gamma_0(\mathfrak{n})] = 144.$$

Proof. By the Hilbert-modular index formula (Shimura [14, Prop. 1.43]),

$$[\mathrm{SL}_2(\mathcal{O}_K) : \Gamma_0(\mathfrak{n})] = N(\mathfrak{n}) \prod_{\mathfrak{p}|\mathfrak{n}} (1 + N(\mathfrak{p})^{-1}).$$

For $K = \mathbb{Q}(\sqrt{5})$ the prime 11 splits as $11 = \pi\bar{\pi}$ with $\pi = (11 + \sqrt{5})/2 \cdot (\text{unit})$ since $11 \equiv 1 \pmod{5}$ and 5 is a quadratic residue mod 11. Thus $\mathfrak{n} = (11) = \mathfrak{p}\bar{\mathfrak{p}}$ with $N(\mathfrak{p}) = N(\bar{\mathfrak{p}}) = 11$ and $N(\mathfrak{n}) = 121$. The formula evaluates to

$$121 \cdot \left(1 + \frac{1}{11}\right) \left(1 + \frac{1}{11}\right) = 121 \cdot \frac{12}{11} \cdot \frac{12}{11} = 144. \quad \square$$

A.2 Containment $\Gamma_\varphi^{(2)} \subseteq \Gamma_0((11))$

Lemma A.2 (Generator containment). *Each squared involution T_{ij}^2 in Γ_φ has lower-left entry in the ideal $(11) \subset \mathbb{Z}[\varphi]$:*

$$c(T_{12}^2) = 0, \quad c(T_{13}^2) = -11, \quad c(T_{23}^2) = 0.$$

Proof. By direct computation from the explicit matrix presentation of T_{ij}^2 (certified by `TestGamma0Level11Inclusion` in `test_selberg.py`). The values $\{0, -11, 0\}$ all lie in $11\mathbb{Z}[\varphi] \subset (11)$. $\square$

Proposition A.3 (Subgroup containment). $\Gamma_\varphi^{(2)} \subseteq \Gamma_0((11))$.

Proof. By definition $\Gamma_0((11)) = \{M \in \mathrm{SL}_2(\mathbb{Z}[\varphi]) : c(M) \in (11)\}$. Lemma A.2 gives $c \in (11)$ for each generator of $\Gamma_\varphi^{(2)}$. Closure of $\Gamma_0((11))$ under matrix multiplication is immediate from the standard c -multiplication rule

$$c(MN) = c(M)d(N) + d(M)c(N),$$

which lies in (11) whenever both $c(M)$ and $c(N)$ do. Hence every word in the generators is in $\Gamma_0((11))$, and $\Gamma_\varphi^{(2)} \subseteq \Gamma_0((11))$. $\square$

A.3 Fibonacci-square uniqueness

Theorem A.4 (Cohn 1964). *The only n for which F_n is a perfect square are $n \in \{0, 1, 2, 12\}$, giving $F_n \in \{0, 1, 1, 144\}$.*

Proof. Cohn [13]; verified computationally up to F_{100} in `TestFibonacciSquareLockIn::test_only_square_fi` $\square$

Corollary A.5 (Codimension-Fibonacci identity). $[\Gamma_\varphi : \Gamma_0((11))] = 144 = F_{12} = (L_5 + 1)^2$.

Proof. The first equality is Proposition A.1. The second is Theorem A.4 ($F_{12} = 144$). The third uses $L_5 = \varphi^5 + \psi^5 = 11$ (the defining Lucas identity for our Ogg prime). $\square$

A.4 Dimension of the weight-2 new cuspform space at level 11

Proposition A.6 (Genus formula). *The modular curve $X_0(11)$ has genus 1. Equivalently,*

$$\dim S_2(\Gamma_0(11))^{\text{new}} = 1.$$

Proof. Apply the genus formula [16, Thm. 3.1.1]:

$$g(X_0(N)) = 1 + \frac{[\text{SL}_2(\mathbb{Z}) : \Gamma_0(N)]}{12} - \frac{\nu_2(N)}{4} - \frac{\nu_3(N)}{3} - \frac{\nu_\infty(N)}{2}.$$

For $N = 11$ prime: $[\text{SL}_2(\mathbb{Z}) : \Gamma_0(11)] = 12$, $\nu_2(11) = 1 + \left(\frac{-1}{11}\right) = 0$, $\nu_3(11) = 1 + \left(\frac{-3}{11}\right) = 0$, $\nu_\infty(11) = 2$. Substituting:

$$g(X_0(11)) = 1 + \frac{12}{12} - 0 - 0 - \frac{2}{2} = 1 + 1 - 1 = 1.$$

The newform identification $\dim S_2(\Gamma_0(11))^{\text{new}} = g(X_0(11))$ is standard (Mazur [17]; Cremona [18]); the unique newform is the Hecke eigenform $f_{E_{11}}$ associated to the elliptic curve E_{11} . $\square$

A.5 Hecke arithmetic at level (11)

Theorem A.7 (Base-change Hecke at level (11)). *Let $f_{E_{11}} \in S_{(2,2)}(\Gamma_0((11)))$ be the Hilbert modular newform arising from $E_{11}/\mathbb{Q}(\sqrt{5})$ by base change. For each rational prime p let $a_p = a_p(E_{11})$ be the p -th Hecke eigenvalue of the weight-2 newform of $\Gamma_0(11)$. Then for every prime ideal $\mathfrak{p} \subset \mathbb{Z}[\varphi]$:*

$$a_{\mathfrak{p}}(f_{E_{11}}) = \begin{cases} a_p & \text{if } p \text{ splits in } \mathbb{Z}[\varphi], \mathfrak{p} \mid p, N\mathfrak{p} = p, \\ a_p^2 - 2p & \text{if } p \text{ is inert, } N\mathfrak{p} = p^2, \\ a_5 = 1 & \text{if } p = 5 \text{ ramified, } N\mathfrak{p} = 5, \\ a_{11} = 1 & \text{if } p = 11 \text{ bad split multiplicative.} \end{cases}$$

Moreover each $a_{\mathfrak{p}}$ satisfies the Ramanujan–Petersson bound $|a_{\mathfrak{p}}| \leq 2\sqrt{N\mathfrak{p}}$.

Proof. Existence of the Hilbert newform $f_{E_{11}}$ is the modularity theorem of Freitas–Le Hung–Siksek [12], which states that every elliptic curve over $\mathbb{Q}(\sqrt{5})$ is modular. The base-change relation $L(f_{E_{11}}, s) = L(E_{11}/\mathbb{Q}(\sqrt{5}), s) = L(E_{11}, s) \cdot L(E_{11} \otimes \chi_5, s)$ is the Shimura base-change theorem [15, §4].

For the local Euler factors: at split $p \nmid 11$, the prime p has two ideal factorisations $\mathfrak{p}, \bar{\mathfrak{p}}$ of equal norm p , and each contributes the same factor as $L(E_{11}, s)$ at p , giving $a_{\mathfrak{p}} = a_p$. At inert $p \nmid 11$, the prime ideal has norm p^2 , and the Euler factor of $L(f_{E_{11}}, s)$ at $\mathfrak{p}$ equals the product of Euler factors of $L(E_{11}, s)$ and $L(E_{11} \otimes \chi_5, s)$ at p , which simplifies to

$$a_{\mathfrak{p}} = a_p^2 - 2p$$

by direct multiplication of the local quadratic factors. At $p = 5$ ramified, the single prime ideal has $a_{\mathfrak{p}} = a_5 = 1$ by Atkin–Lehner theory of E_{11} at the good prime 5. At $p = 11$ split multiplicative bad reduction is preserved on base-change since 11 is unramified in $\mathbb{Z}[\varphi]/\mathbb{Q}$; both $\mathfrak{p}, \bar{\mathfrak{p}} \mid 11$ inherit the same eigenvalue $a_{11} = 1$.

The Ramanujan–Petersson bound for $f_{E_{11}}$ as a weight-(2, 2) Hilbert cusp form is Blasius [19] (the Hilbert-modular case of Deligne’s theorem). $\square$

This completes the formal-theorem foundation. Conjecture 5.1 (Selberg-zeta factorisation through $L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$ inside $Z_{\Gamma_0((11))}$) remains open and is the subject of §6 and follow-up work.

A.6 Sato–Tate stratification and the splitting–Peirce identity

Theorem A.7 gives the formula $a_{\mathfrak{p}} = a_p$ at split primes and $a_{\mathfrak{p}} = a_p^2 - 2p$ at inert primes. After Sato–Tate rescaling $x_{\mathfrak{p}} := a_{\mathfrak{p}}/(2\sqrt{N\mathfrak{p}})$, these become

- *split*: $x_{\mathfrak{p}} = x_p$ where x_p obeys the semicircle density $f_{\text{sc}}(x) = \frac{2}{\pi}\sqrt{1-x^2}$ by the Barnet-Lamb–Geraghty–Harris–Taylor proof of Sato–Tate for non-CM $E/\mathbb{Q}$ (BLGHT, 2011);
- *inert*: $x_{\mathfrak{p}} = 2x_p^2 - 1$, the second Chebyshev pushforward of semicircle, with density $f_{\text{inert}}(x) = \frac{1}{\pi}\sqrt{(1-x)/(1+x)}$ on $(-1, 1]$;
- *ramified*: the single value $1/(2\sqrt{5})$.

This is the literal content of the *splitting–Peirce identity* of the metabolic paper (§ “The Spectral Reading” therein): the three Sato–Tate strata at the three splitting types of $\mathbb{Z}[\varphi]$ correspond, density by density, to the three Peirce sectors $P_4/P_{4c}/P_{23}$ of the $\text{Cl}(3, 1)$ algebra used in the genesis dynamics.

Theorem A.8 (Empirical universality across non-CM curves). *For each of 10 non-CM elliptic curves $E/\mathbb{Q}$ of small conductor coprime to 5 (specifically $\{11a1, 14a1, 17a1, 19a1, 21a1, 26a1, 33a1, 37a1, 43a1, 53a1\}$) the rescaled Hecke eigenvalues $\{x_{\mathfrak{p}}\}$ at primes $\mathfrak{p}$ of $\mathbb{Z}[\varphi]$ above rational primes $p \leq 4000$ pass the two-sided Kolmogorov–Smirnov test against f_{sc} on the split stratum and against f_{inert} on the inert stratum at significance $\alpha = 0.01$. Each stratum fits its predicted density better than the alternative, with the wrong-density K – S statistic exceeding the correct-density K – S by a factor of ≥ 4 . All 4 CM curves ($\{27a1, 32a1, 36a1, 49a1\}$, with CM by $\mathbb{Q}(\sqrt{-3}), \mathbb{Q}(i), \mathbb{Q}(\sqrt{-3}), \mathbb{Q}(\sqrt{-7})$ respectively) fail the test with K – S statistics ≥ 3 times the critical value and supersingular fractions of $\sim 34\%$, within 16% of the Chebotarev asymptote $1/2$.*

Proof (empirical). PARI/GP computes $a_{\mathfrak{p}}$ for each $(E, \mathfrak{p})$ via the base-changed `ellap`; the Python module `substrate-sim/algebra/arithmetric_sato_tate.py` performs the stratified K – S test. All 52 assertions (per-curve $\times$ per-stratum + population-level effect-size checks) pass at HEAD. $\square$

Theorem A.9 (Cross-field invariance). *The conclusion of Theorem A.8 holds with $K = \mathbb{Q}(\sqrt{5})$ replaced by $\mathbb{Q}(\sqrt{2})$ or $\mathbb{Q}(\sqrt{13})$, at $p \leq 8000$. For each of the three real quadratic fields, all 10 non-CM curves pass the K -S falsifier and all 4 CM curves fail. The Peirce labelling $P_4/P_{4c}/P_{23}$ of the splitting types ramified / inert / split is therefore K -independent: it is a property of the splitting structure of the abstract real quadratic field, not of $\mathbb{Q}(\sqrt{5})$ specifically.*

Remark A.10. Theorems A.8–A.9 are stated as empirical theorems because they are existence statements about the K -S statistics on a specified finite sample. They do not upgrade the underlying Sato–Tate equidistribution to a stronger form. Their content is the structural identification of the three arithmetic strata with the three algebraic Peirce sectors, verified at 42 different (E, K) pairs.

Companion engineering artefacts

- `substrate-sim/algebra/apollonian.py` — classical integer gasket checks.
- `substrate-sim/algebra/apollonian_zphi.py` — Vieta & norm series generators.
- `substrate-sim/algebra/moonshine.py` — partitioning & comparison harness.
- `substrate-sim/algebra/selberg.py` — SSA matrix reduction, congruence theorem, closed geodesic enumeration with *exact* $\mathbb{Z}[\varphi]$ -arithmetic on $\mathbb{H} \times \mathbb{H}$, partial Hilbert-Selberg log-derivative, $L(E_{11})$ and base-change $L(E_{11}/\mathbb{Q}(\sqrt{5}), s) = L(E_{11}) \cdot L(E_{11} \otimes \chi_5)$ Euler-product log-derivatives.
- `substrate-sim/algebra/run_spectral_test.py` — spectral comparison runner: enumerates double-hyperbolic geodesics of $\Gamma_\varphi^{(2)}$ up to word length 10, computes the partial Hilbert-Selberg log-derivative versus $-L'/L(E_{11}/\mathbb{Q}(\sqrt{5}), s)$, and tabulates the decay-ratio convergence $\log_2(\text{geo})/\log_2(L) \rightarrow 20 \log_5 \varphi \approx 5.98$.
- `substrate-sim/algebra/tests/test_apollonian*.py` — negative coefficient-matching oracle.
- `substrate-sim/algebra/tests/test_selberg.py` — 72 passing tests for the level-11 structure: congruence, SSA at $p \in \{19, 29, 31, 41\}$, Galois trace -11^3 , Hecke eigenvalues, the fundamental norm identity $N_{\text{gal}}(T_{12}^2) = \varphi^{20}$, the *spectral-scale-mismatch* certification (6 assertions in `TestSpectralScaleMismatch`) fixing the exact algebraic scale-ratio $20 \log_5 \varphi$, the *codimension lock-in* (`TestGamma0Level11Inclusion`, 12 assertions) proving $\Gamma_\varphi^{(2)} \subseteq \Gamma_0((11))$ at index 144, and the *Fibonacci-square lock-in* (`TestFibonacciSquareLockIn`, 9 assertions) showing $144 = F_{12}$ is the unique non-trivial square Fibonacci by Cohn (1964).
- `substrate-sim/algebra/gamma0_11.py` — $\Gamma_0((11))$ generators (extending $\Gamma_\varphi^{(2)}$ by T_1, T_φ, U_{11}), splitting types {split, inert, ramified, bad} in $\mathbb{Z}[\varphi]$, base-change Hecke-eigenvalue predictions $a_p(f_{E_{11}})$, Ramanujan-Petersson bound, Hilbert L-function Dirichlet-coefficient recovery.
- `substrate-sim/algebra/tests/test_gamma0_11.py` — 49 passing tests certifying the Hecke arithmetic of §6: split/inert/ramified classification at primes ≤ 97 , predicted a_p table including the flagship $a_p = 0$ at smallest split prime $p \mid 19$, Ramanujan bound, inert-prime p^{-s} cancellation in $L(E_{11}/\mathbb{Q}(\sqrt{5}))$, and the scale separation from the ramified prime $p = 5$.
- `substrate-sim/algebra/onan_moonshine.py` — the six pariah sporadic groups with orders and prime factorisations, $|O'N|$ verification, pariah classes A and B, initial coefficients $A_e(D)$ of the O’Nan moonshine identity series with $A_e(-11) = 1$, Shimura-correspondence target prediction $\text{Sh}(Z_e) \propto f_{E_{11}}$ at level 11.

- `substrate-sim/algebra/tests/test_onan_moonshine.py` — 38 passing tests certifying: exactly 6 pariah groups, $|O'N| = 460\,815\,505\,920$ and its factorisation, pariahs containing $p = 11$ ($J_1, J_4, Ly, O'N$), Class A vs B partition (J_1 and $O'N$ are Class A), small O'Nan moonshine coefficients including $A_e(-11) = 1$, Shimura target dimension 1 at level 11 (= genus of $X_0(11)$), and the pariah-count = spectral-immersion-depth agreement to 1%.
- `substrate-sim/algebra/tests/test_hopf_seam.py` — 41 passing tests certifying the Hopf-seam framework: nine Heegner numbers and CM j -invariants; $j(E_{11}) = -4096/11$ is not a CM j -invariant; minimal polynomials $\varphi^2 - \varphi - 1 = 0$ and $\omega_{-11}^2 - \omega_{-11} + 3 = 0$ at the same prime; Heegner $\cap$ Ogg = $\{2, 3, 7, 11, 19\}$; pariah Class A vs Class B partition; the five-fold convergence at $p = 11$.
- `substrate-sim/algebra/tests/test_minimality.py` — 27 passing tests certifying Propositions 7.2 and 7.3: the $(\mathbb{Q}(\sqrt{5}), \varphi, k = 5, p = 11)$ construction is the unique smallest-discriminant real-quadratic thin-group construction with non-trivial Lucas recurrence reaching an Ogg prime with positive-genus modular curve.
- `substrate-sim/algebra/hilbert_trace.py` (Phase 3 β) — Hilbert-modular dimension formula at level (11) giving $\dim S_{(2,2)}(\Gamma_0((11))) = 10$ (of which one is the base-change of $f_{E_{11}}$ and nine are genuinely-new Hilbert newforms), geometric/spectral partial-sum trace formula, and the factorisation residual at $s \in \{3, \dots, 10\}$.
- `substrate-sim/algebra/tests/test_geodesic_hecke.py` (Phase 3 α) — 55 tests certifying: $\sqrt{5} \bmod p$ Tonelli–Shanks at split primes; Frobenius reduction at $\mathfrak{p}$ with the consistency relation $t \cdot \bar{t} \equiv N_{K/\mathbb{Q}}(a + b\varphi) \pmod{p}$; chi-squared discrimination between $a_{\mathfrak{p}} = 0$ primes ($\mathfrak{p} \mid 19, \mathfrak{p} \mid 29$, normalised $\chi^2 < 1$) and $|a_{\mathfrak{p}}| > 5$ primes ($\mathfrak{p} \mid 31, \mathfrak{p} \mid 41$, normalised $\chi^2 > 1$).
- `substrate-sim/algebra/pari_crossvalidate.py` (Phase 3 ε) — PARI/GP-based verification of every $a_p(E_{11})$ at primes ≤ 97 . Found and fixed eight previously incorrect entries in the hardcoded table.
- `substrate-sim/algebra/pariah_depth.py` (Phase 3 γ) — Three independent structural attempts at $6 = \lceil 20 \log_5 \varphi \rceil$ (weight-1 cusp forms, Niemeier-Ogg lattice count, Hopf Chern class). All three fail to give a structural identification; the agreement is downgraded to “intriguing 0.4% coincidence.”

Test suite totals after Phase 3. 741 passing tests across the algebra suite (up from 586 post-Phase-2): Phase 3 added 155 new tests across new modules (`test_hilbert_trace.py` 33, `test_geodesic_hecke.py` 55, `test_pari_crossvalidate.py` 35, `test_pariah_depth.py` 16, `test_phase3_catalog.py` 10, plus 6 new tests in `test_onan_moonshine.py` for the Shimura proportionality theorem). The catalog explicitly maps every Phase 3 theorem T1–T5 and every open conjecture to at least one certifying test method.

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